

The synaptic Package: User Manual

synaptic project

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Abstract: **synaptic** is a modular LaTeX3 design system for journal articles, books, lecture material, and compact notes. This manual describes its installation, requirements, safe configuration model, typographic system, metadata API, theorem system, theme and customization tools, and mode-specific features.

Keywords: synaptic, LaTeX3, LuaLaTeX, academic typesetting

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1 ■ Getting started

The package requires LuaLaTeX and a KOMA-Script class. If the package is installed in your TeX tree, a document needs nothing more than:

```
\documentclass{scrartcl}
\usepackage[mode=journal]{synaptic}
\SynapticTitle{A Short Article}
\SynapticAuthor{Ada Author}
\begin{abstract}Summary.\end{abstract}
\begin{document}
\SynapticMakeTitle
\section{Introduction}
Text.
\end{document}
```

Use `scrbook` for book mode. The selected class controls the paper size; synaptic does not silently force A4.

1.1 Using a checkout without installation

The repository ships generated package files. Point `TEXINPUTS` at them:

```
export TEXINPUTS=/path/to/synaptic/tex/latex/synaptic//:$TEXINPUTS
lualatex document.tex
```

1.2 System installation

A TeX Live installation can install (and, when versions change, replace) the bundle from the repository root:

```
l3build install    # copy into the user's texmf tree
l3build uninstall  # remove a previous installation
l3build ctan       # produce TDS/CTAN archives
```

Any standard `texmf` tree works; the generated files belong in `tex/latex/synaptic/`. Archives from `l3build ctan` unpack directly into the same structure.

1.3 Requirements

- LuaLaTeX; pdfLaTeX and XeLaTeX are rejected with an explicit error.
- A KOMA-Script class (`scrartcl`, `scrreprt`, or `scrbook`).
- A recent TeX distribution with `expl3`, `fontspec`, `unicode-math`, `geometry`, `microtype`, `thmtools`, and the other standard dependencies.
- `tcolorbox` and PGF for teaching and notes cards; `ctex` and a CJK font for `lang=zh`.

2 ■ Configuration

Key	Default	Values and purpose
<code>mode</code>	journal	journal, book, lecture, or notes
<code>theme</code>	ocean	ocean, graphite, forest, midnight, or paper
<code>lang</code>	en	en or zh; selects labels and CJK support
<code>fontset</code>	auto	auto, xcharter, libertine, stix2, or lm
<code>toc</code>	false	whether the title workflow prints a TOC
<code>toc-layout</code>	top	top, inline, or page
<code>title-layout</code>	auto	inline, compact, page, cover, or mode default
<code>numbering</code>	auto	section, chapter, continuous, none, or mode default
<code>box-numbering</code>	shared	shared, separate, or none
<code>color-mode</code>	screen	screen, print, or mono
<code>density</code>	balanced	compact, balanced, or airy
<code>measure</code>	auto	narrow, standard, wide, or mode default
<code>theorem-style</code>	boxed	boxed or plain
<code>binding-offset</code>	0pt	additional book binding correction
<code>extra-math</code>	false	convenience Unicode math alphabets
<code>fine-breaking</code>	true	moderate emergency stretch and tolerance

Only `theme`, `toc`, and `toc-layout` are safe runtime settings:

```
\SynapticSetup{theme=paper,toc=false,toc-layout=page}
```

Mode, language, font, and load-time typography options are frozen after package loading. Attempts to change them issue a warning instead of producing a half-reconfigured document.

2.1 Modes

Mode	Class	Purpose
journal	<code>scrartcl</code>	Inline article title, metadata, running heads, statements
book	<code>scrbook</code>	Book title sequence, parts, chapters, duplex navigation
lecture	<code>scrartcl/scrreprt</code>	Course masthead and semantic teaching system
notes	<code>scrartcl</code>	Compact masthead and lightweight knowledge cards

journal renders an inline article heading on the first page. The running head carries the first author and short title, and the block supports abstract, keywords, arXiv, correspondence, publication information, and statement sections for journal submissions.

book renders a half title, formal title page, verso block, and optional dedication, then provides front/main/back matter helpers, part pages, chapter subtitles, epigraphs, and mirrored duplex running heads.

lecture renders a course masthead (course code, title, lecturer, semester, lecture number) and wraps theorems and examples in a sharp card system. Learning goals and lecture summaries frame each teaching unit.

notes renders a compact masthead with an optional tag line and supports class-level two-column composition with balanced final columns.

2.2 Fonts

The **auto** setting chooses the first *complete* installed bundle: XCharter, Libertinus, STIX Two, then Latin Modern. An explicit choice reports a clear error if a required companion font is absent. Chinese documents prefer Noto Serif/Sans CJK SC and fall back to the Fandol fonts distributed with TeX Live.

3 ■ The typographic system

Synaptic’s default appearance is deterministic rather than decorative. A handful of design decisions repeat throughout every mode.

3.1 Colour delivery

color-mode	Use	Effect
screen	on-screen reading	full theme colours, coloured links
print	paper output	theme colours, dark (print-safe) links
mono	monochrome delivery	black-grey tokens only; semantic colours map to greys

3.2 Density and measure

density tunes line spacing in three steps: **compact** for dense reference material, **balanced** as the mode-aware default, and **airy** for slide-like handouts. **measure** overrides the mode-default column width: **narrow** for long prose, **standard** for journals, **wide** for design-driven layouts. In book mode, measure adjusts the mirrored inner/outer margins instead of breaking duplex symmetry.

3.3 Title composition

title-layout selects how each mode renders its title area. The **auto** value resolves to **inline** (journal), **cover** (book), and **compact** (lecture, notes). **page** dedicates a full title page and **inline** forces the first-page composition. All title renderers reuse an asymmetric “synaptic rail”: a short primary rule, an accent node, and a quiet continuation line. Book title and verso pages use the motif at architectural edges; lecture and notes place it beside compact metadata panels, while journal mode keeps the treatment editorial and inline.

3.4 Navigation

Every mode draws a hairline beneath its running heads. The hairline uses a separate, deliberately lighter decorative token so navigation stays quiet. Running-head labels and outer folios repeat the small node marker; book mode additionally gives forced duplex blank pages a quiet footer anchor.

3.5 Headings, contents, and captions

All numbered section levels share the sans heading family: ‘section’ uses the primary colour, ‘subsection’ the dark text colour, and ‘subsubsection’ a smaller, slightly muted tint, so structure reads at a glance. When a table of contents is requested its top-level entries repeat the on-page heading colour in bold while deeper entries stay dark, and the number alignment and vertical rhythm are tightened. Figure and table caption labels carry the secondary theme token to keep them consistent with the rest of the design.

3.6 Cards and rules

Teaching and notes cards share one visual language: a very small corner radius, a west accent bar tinted with the semantic colour of the card (primary, secondary, warning, or success), a faint hairline, and a light surface with a readable title colouring. Rules and hairlines derive from the theme’s primary colour at low tint, so decoration never competes with text.

4 ■ Title and PDF metadata

All four modes share the metadata model and render it differently:

Command	Meaning
<code>\SynapticTitle</code>	Required printed title and PDF title
<code>\SynapticSubtitle</code>	Optional display subtitle
<code>\SynapticShortTitle</code>	Running-head title
<code>\SynapticDate</code>	Explicit date
<code>\SynapticAuthor</code>	Repeatable author and optional affiliation mark
<code>\SynapticAffiliation</code>	Repeatable affiliation
<code>\SynapticEmail</code>	Repeatable clickable contact address
<code>\SynapticSubject</code>	PDF subject
<code>\SynapticKeywords</code>	Printed and PDF keywords
<code>\SynapticHeader</code>	Submission header override
<code>\SynapticSubHeader</code>	Secondary header
<code>\SynapticPreprint</code>	Preprint identifier
<code>\SynapticArxiv</code>	Linked arXiv identifier
<code>\SynapticDedicated</code>	Dedication
<code>\SynapticMakeTitle</code>	Validate and render all title data

Journal mode accepts publication information, correspondence, and author notes; book mode accepts half-title, publisher, edition, ISBN, and copyright fields; lecture mode accepts course, code, lecture number, instructor, and semester. Use `\SynapticStatement{heading}{body}` for journal funding, data, ethics, or conflict statements; the starred form omits its table-of-contents entry. Standard `\title`, `\author`, and `\date` are imported when the Synaptic field is empty. PDF metadata receives one Unicode-safe conversion.

5 ■ Theorems and mathematics

The built-in `theorem`, `lemma`, `proposition`, `definition`, `corollary`, `axiom`, `conjecture`, `claim`, `property`, `criterion`, `convention`, `problem`, `solution`, `example`, `remark`, and `note` environments share one counter. Books use chapter scope by default; shorter modes use sections. Definitions use upright body text; proposition-style statements use italics in English and upright CJK text.

By default (`theorem-style=boxed`) the numbered statement environments are presented as refined cards: a subtle theme-tinted background, a semantic west accent bar, a faint hairline, and the same restrained corner treatment as the teaching and notes card language. `proof` stays clean and unboxed. Select `theorem-style=plain` for the classic unboxed look.

Theorem 5.1 (Gaussian integral). *The integral satisfies*

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

Proof. Square the integral, change to polar coordinates, and evaluate the radial integral. ■

New theorem types accept a style:

```
\SynapticNewTheorem[remark]{observation}{Observation}
```

The available styles are `plain`, `definition`, and `remark`; a custom theorem is boxed automatically. Repeating a declaration for an existing environment is safe: synaptic keeps the established definition and emits a warning, which lets older documents coexist with the expanded built-in theorem set.

Observation 5.2. With `extra-math=true`, commands such as $\mathbb{R}$, $\mathcal{F}$, and $\mathfrak{g}$ are available.

6 ■ Themes and customization

Themes may be switched with `\SynapticUseTheme{name}`. A custom theme is defined and activated using `xcolor` expressions:

```
\SynapticDefineTheme{brand}{blue!70!black}{gray!80!black}[orange]
```

The definition arguments are primary, secondary, and optional accent colours. The semantic tokens below are always available for extensions and always derive from the active theme:

Token	Role	Use
<code>syn-primary</code>	brand colour	headings, title rules, accents
<code>syn-secondary</code>	ink colour	muted text, remark tones
<code>syn-accent</code>	accent colour	optional emphasis
<code>syn-text-dark</code>	body ink	readable text
<code>syn-muted-text</code>	readable muted	auxiliary labels
<code>syn-surface</code>	card background	boxes and surfaces
<code>syn-border</code>	card border	box frames
<code>syn-rule</code>	decorative hairline	separators
<code>syn-danger</code>	semantic danger	warnings on the red side
<code>syn-warning</code>	semantic warning	warnings on the amber side
<code>syn-success</code>	semantic success	exercises, todo lists

Lecture mode adds collision-resistant teaching environments. Each takes an optional title and an optional label:

- `synexample` — worked examples, share the theorem counter by default
- `synremark` — remarks, unnumbered by default
- `synwarning` — warnings, unnumbered by default
- `synexercise` — exercises, share the theorem counter
- `synbox` — generic box with an optional title

The generic v2.0 aliases are created only when their names are free. Course units may also use `synlearninggoals` and `synlecturesummary`.

7 ■ Book workflow and acknowledgments

Use `\SynapticFrontMatter`, `\SynapticMainMatter`, and `\SynapticBackMatter` around the corresponding parts of a book. `\SynapticBookContents`, chapter subtitles, epigraphs, and list/appendix helpers complete the long-form workflow. Notes mode provides `synkeyidea`, `synquestion`, `syntodo`, and `synsummary`; class-level two-column layouts are supported. The `\SynapticAcknowledgments` command creates a chapter in book mode and a section in other modes. Its starred form suppresses the TOC entry.

8 ■ Troubleshooting

- “The synaptic package requires LuaLaTeX” — you are running pdfLaTeX or XeLaTeX. Engines other than LuaLaTeX are rejected on purpose.
- “requires a KOMA-Script document class” — load the package with `scrartcl`, `scrreprt`, or `scrbook`.
- “mode=book requires a chapter-based class” — use `scrbook` (or another KOMA class defining `chapter` and `frontmatter`).

- **“Incomplete XCharter/Libertinus/STIX Two font bundle”** — the explicit `fontset` is missing a companion font. Install the bundle, or use `fontset=auto` to fall back automatically.
- **Options frozen after loading** — `mode`, `lang`, `fontset`, and `typography` keys are structural. Re-select them in `\usepackage`; do not attempt runtime changes.
- **No CJK font found** — install Noto CJK SC or the Fandol fonts used as the TeX Live fallback.
- **“Document title is empty”** — set `\SynapticTitle` (or `\title`) before calling `\SynapticMakeTitle`.
- **Paper size unwanted** — the package never overrides the class paper size; select it in the class options.

9 ■ Migrating from earlier versions

Version 2.x replaced the `v1` option and command surface. The removed `style=` and `color=` options remain explicit errors with migration hints.

v1	v2.5
<code>color=ocean</code>	<code>theme=ocean</code>
<code>style=modern/minimal</code>	<code>mode=journal/notes</code>
<code>\title</code>	<code>\SynapticTitle</code>
<code>\author</code>	<code>\SynapticAuthor</code>
<code>\maketitle</code>	<code>\SynapticMakeTitle</code>

The v2.5 API remains compatible with v2.2. Lower-case book matter helpers and generic lecture box names remain conditional v2.0 compatibility aliases; new documents should use the namespaced public forms.

10 ■ Development commands

```
l3build unpack  # regenerate .sty and .def files from synaptic.dtx
l3build check   # run the LuaLaTeX regression suite (18 groups)
l3build doc     # build the documented source and manuals
l3build ctan    # create release archives
```

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